THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET D

Chapters 7 & 8: **Proportional Reasoning-1, Fractions in Disguise** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ratio of 1 m to 40 cm is: 1

(a) 5 : 2 **(b)** 1 : 40 **(c)** 2 : 5 **(d)** 25 : 1

2. If the cost of 6 pens is ₹ 90, the cost of 10 pens is: 1

(a) ₹ 100 **(b)** ₹ 180 **(c)** ₹ 120 **(d)** ₹ 150

3. If A : B = 2 : 3 and B : C = 4 : 5, then A : B : C is: 1

(a) 8 : 12 : 15 **(b)** 2 : 4 : 5 **(c)** 2 : 3 : 5 **(d)** 6 : 8 : 10

4. The ratio of 250 g to 2 kg is: 1

(a) 1 : 80 **(b)** 1 : 4 **(c)** 1 : 8 **(d)** 5 : 4

5. The cost of 12 eggs is ₹ 90. The cost of 18 eggs is: 1

(a) ₹ 108 **(b)** ₹ 120 **(c)** ₹ 150 **(d)** ₹ 135

6. The simple interest on ₹ 5000 at 8% per annum for 2 year(s) is: 1

(a) ₹ 1600 **(b)** ₹ 800 **(c)** ₹ 400 **(d)** ₹ 5800

7. The simple interest on ₹ 15000 at 6% per annum for 4 year(s) is: 1

(a) ₹ 18600 **(b)** ₹ 1800 **(c)** ₹ 7200 **(d)** ₹ 3600

8. The compound interest on a sum for 2 years is greater than the simple interest because: 1

(a) principal increases at start
(b) rate is higher
(c) time is longer
(d) interest is also earned on earlier interest

9. GST is a tax charged on: 1

(a) goods and services **(b)** income only
(c) property transfer only **(d)** land only

10. ₹ 1000 at 10% compound interest annually for 2 years gives an amount of: 1

(a) ₹ 1100 **(b)** ₹ 1220 **(c)** ₹ 1210 **(d)** ₹ 1200

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write the ratio 45 minutes : 1 day in simplest form. 2

12. What is direct proportion? Give an example. 2

13. The cost of an item with 18% GST on ₹ 500 is? 2

14. Find the discount when SP is ₹ 540 and MP is ₹ 600. Express it as a percentage. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A train travels 150 km in 2.5 hours. At the same speed, find the time to cover 270 km. 3

16. A bicycle bought for ₹ 4000 is sold at a loss of 12.5%. Find the SP. 3

17. A mixture of 40 L contains milk and water in the ratio 7 : 1. How much water must be added to make the ratio 7 : 3? 3

OR

The perimeters of two squares are in the ratio 2 : 3. Find the ratio of their sides and areas. 3

18. A man saves ₹ 4500 per month, which is 15% of his salary. Find his salary and annual savings. 3

OR

Find the rate of simple interest if ₹ 8000 earns ₹ 1920 interest in 3 years. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) The ratio of incomes of two persons is 5 : 4 and expenses 4 : 3. If each saves ₹ 1000, find incomes. 5
(ii) Verify.

OR

(i) Find the ratio of the areas of two squares with sides 7 cm and 14 cm. 5
(ii) Compare with the ratio of the sides.

20. (i) Express the ratio 3 : 5 as percentages. 5
(ii) A mixture is 35% acid; how much acid in 80 L?

OR

(i) A phone bought for ₹ 15000 loses 20% value in the first year and 10% in the second. Find its value after 2 years. 5
(ii) Total depreciation percent.

END OF QUESTION PAPER — Total: 40 Marks

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